

Tshana Tumelo Konaite

Software Engineer

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Personal Statement

I am an independent and goal-driven Data Scientist with a strong academic background in Computational and Applied Mathematics and over six years of professional experience. I possess proven research, analytical and problem-solving skills, with the ability to translate complex concepts into practical, data-driven solutions. My experience across machine learning, software development and cloud technologies enables me to contribute significant value in high-performance environments, while my commitment to continuous learning positions me for continued professional growth.

CORE SKILLS & TECHNOLOGY STACK

Languages: Python, SQL, JavaScript, HTML5, CSS3, Bash

Backend: FastAPI, Flask, SQLAlchemy, Pydantic

AI and Machine Learning: Generative AI, LLM integration, RAG, prompt engineering, TensorFlow, Scikit-learn, Pandas, NumPy, FAISS, ChromaDB

Security: Keycloak, OAuth 2.0, JWT, RBAC, Firebase

Data Technologies: PostgreSQL, MySQL, Redis, DynamoDB, MongoDB

Cloud and DevOps: AWS, Huawei Cloud, Docker, GitHub Actions, Terraform, Ansible, CI/CD

APIs and Messaging: REST APIs, Kafka, webhooks, OpenAPI,

Testing and Tools: Pytest, unittest, Linux, Git, Postman, Swagger, Bash scripting

PROFESSIONAL Experience

Software Engineer

Beautyverse

Dec 2025 – Present

Core Responsibilities:

Develop backend and data services for BeautyVerse's business management platform and marketplace using Python, FastAPI, SQLAlchemy, and PostgreSQL. Design structured database models, data validation workflows, REST APIs, and external service integrations while ensuring data integrity, reliability, and maintainability. Support automated testing, deployment, and infrastructure processes using AWS, Docker, Linux, CI/CD pipelines, and Infrastructure as Code.

Data Analyst

Huawei Technologies

Dec 2019 – Nov 2025

Core Responsibilities:

A data analyst responsible for developing and maintaining centralised data pipelines, automation frameworks, and cloud-based reporting solutions for large-scale telecommunications projects. Used Python, SQL, Huawei Cloud, and Linux to cleanse, validate, reconcile, and analyse operational data, improving reporting accuracy, auditability, forecasting, and decision-making. Applied quantitative analysis and automated workflows to support network planning, delivery performance, cost management, and operational efficiency.

PROFESSIONAL CERTIFICATIONS

2026 – AWS Certified Generative AI Developer – Professional (Early Adopter)

2026 – AWS Certified Solutions Architect – Associate

2026 – SAS: Certified Associate – Programming Fundamentals Using SAS 9.4

2025 – AWS Certified Machine Learning Engineer – Associate

2025 – AWS Certified AI Practitioner

EDUCATION

University of the Witwatersrand

Feb 2021 – Nov 2024

MSc in Computational and Applied Mathematics

Completed an advanced postgraduate programme specialising in mathematical modelling, stochastic calculus, numerical methods, and the pricing of interest-rate derivatives. The research compared the LIBOR Market Model and Generalised Forward Market Model, including the modelling of forward- and backward-looking risk-free rates and the valuation of caps and swaptions.

University of the Witwatersrand

Feb 2019 – Nov 2019

BSc Honours in Mathematical Sciences

Completed an interdisciplinary programme focused on advanced mathematics, statistics, optimisation, quantitative finance, operations research, and computational techniques for solving complex business and industry problems.

SELECTED PROJECTS

KasiFPL | [Link](#)

Built an AI-assisted platform that retrieves and analyses Fantasy Premier League expert content from online videos. The system processes transcripts, extracts recommendations, compares expert opinions, generates consensus insights, and presents gameweek-specific recommendations through a web dashboard.

BeautyVerse Business Management Platform | [Link](#)

Developed backend APIs and data services for a multi-tenant business management and marketplace platform. Implemented user management, role-based permissions, provider onboarding, bookings, reviews, media uploads, notifications, and Keycloak authentication using FastAPI, SQLAlchemy, PostgreSQL, Docker, and AWS.